Scenario No: 1 | Linux Command Line Operations Assessment

Task Objective: Configure a new isolated Linux user environment for a trainee analyst named 'john' by establishing proper directory hierarchies, creating a system information script, managing permissions, and granting required administrative privileges via the terminal.

◻ Step 1: Directory Structure Creation

Requirement: Create a directory named Final_exam. Inside Final_exam, create the following directories: (reports, tools, Backup, notes).

Executed Terminal Commands:

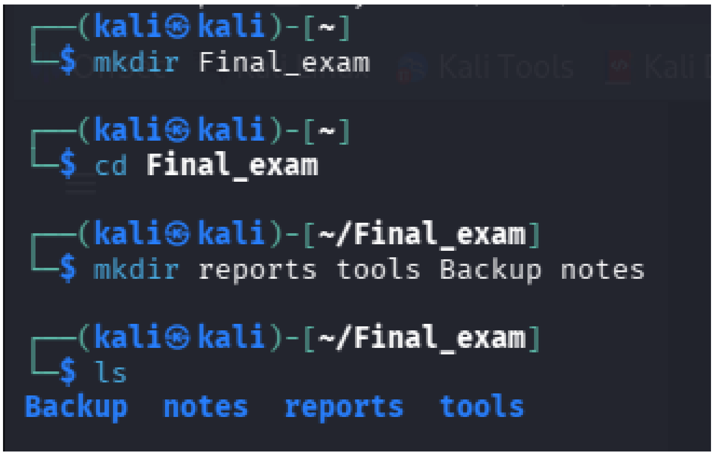
$ mkdir Final_exam

$ cd Final_exam

$ mkdir reports tools Backup notes

$ ls

Technical Explanation: The mkdir Final_exam command initializes the root directory for this assessment. Navigating into it via the change directory (cd) command, the secondary multi-directory creation syntax builds all targeted workspaces (reports, tools, Backup, notes) efficiently in a single execution line. The ls command confirms successful validation of the standard folder architecture.



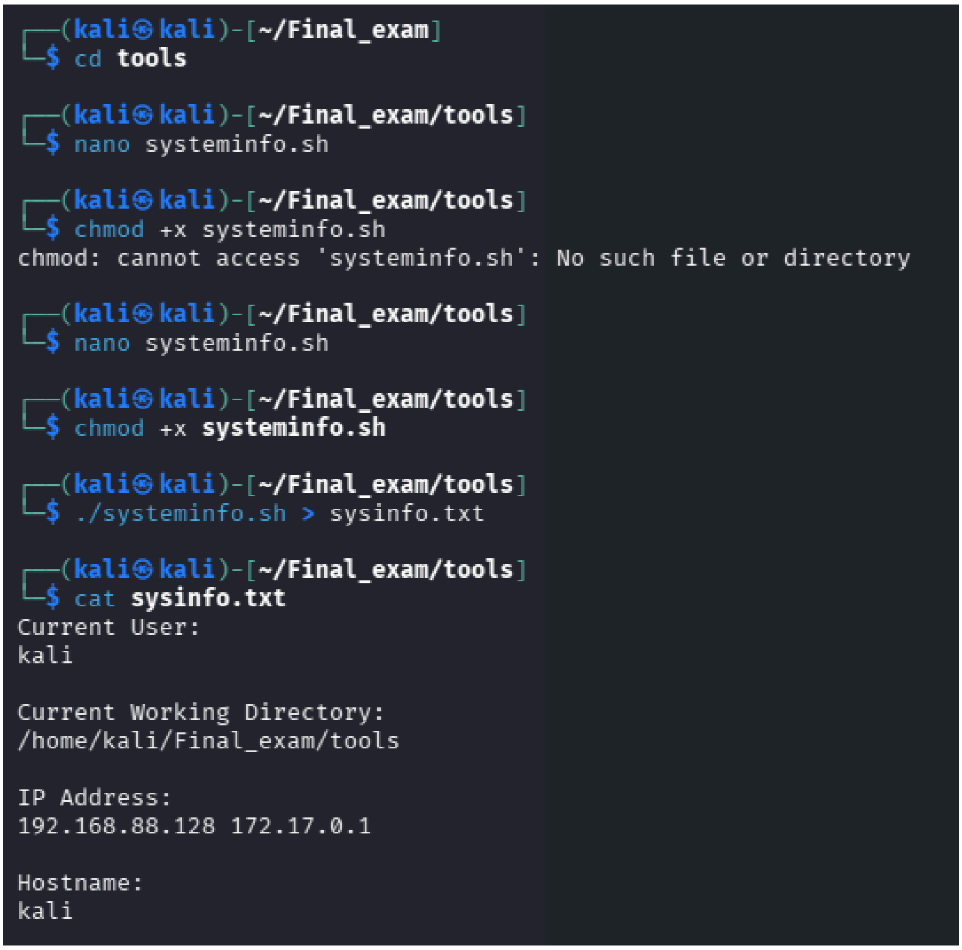
◻ Step 2: Script Creation & Output Generation

Requirement: Inside the tools directory, create a script that prints the following information: Current logged-in username, Current working directory, IP address information, Hostname. Execute the script and save the output into a file named sysinfo.txt.

Executed Terminal Commands:

```
$ cd tools
$ nano systeminfo.sh


# ---- Inside the systeminfo.sh script text editor ----
#!/bin/bash
echo "Current User:"
whoami
echo "Current Working Directory:"
pwd
echo "IP Address:"
hostname -I
echo "Hostname:"
hostname
# ------------------------------------------------


$ chmod +x systeminfo.sh
$ ./systeminfo.sh > sysinfo.txt
$ cat sysinfo.txt
```

Technical Explanation: The automation script systeminfo.sh utilizes system core binaries (whoami, pwd, hostname -I, and hostname) to pull real-time environment status. Executing chmod +x assigns dynamic execution rights to the file system. The standard output redirection operator > captures the terminal sequence stream directly into sysinfo.txt, which is instantly verified and outputted to screen via cat.



```
┌──(kali㉿kali)-[~/Final_exam]
└─$ cd tools

┌──(kali㉿kali)-[~/Final_exam/tools]
└─$ nano systeminfo.sh

┌──(kali㉿kali)-[~/Final_exam/tools]
└─$ chmod +x systeminfo.sh
chmod: cannot access 'systeminfo.sh': No such file or directory

┌──(kali㉿kali)-[~/Final_exam/tools]
└─$ nano systeminfo.sh

┌──(kali㉿kali)-[~/Final_exam/tools]
└─$ chmod +x systeminfo.sh

┌──(kali㉿kali)-[~/Final_exam/tools]
└─$ ./systeminfo.sh > sysinfo.txt

┌──(kali㉿kali)-[~/Final_exam/tools]
└─$ cat sysinfo.txt
Current User:
kali

Current Working Directory:
/home/kali/Final_exam/tools

IP Address:
192.168.88.128 172.17.0.1

Hostname:
kali
```

◻ Step 3: File Management & Archiving

Requirement: Move the sysinfo.txt file into the backup directory and rename it to backup_of_sysinfo.txt.

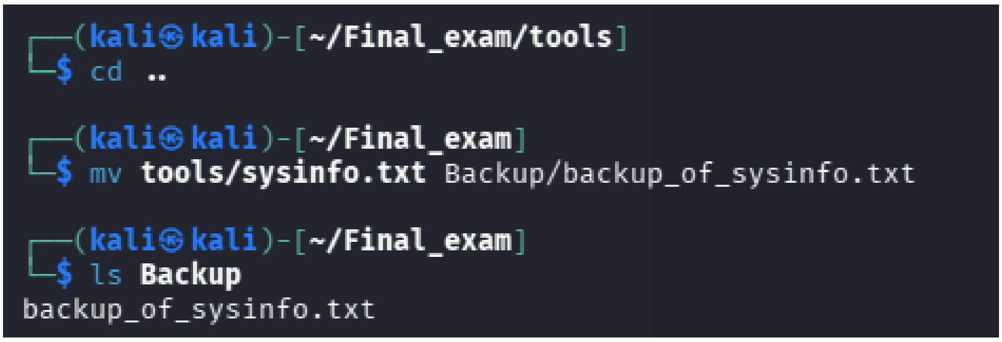
Executed Terminal Commands:

$ cd ..

$ mv tools/sysinfo.txt Backup/backup_of_sysinfo.txt

$ ls Backup

Technical Explanation: Navigating back to the root workspace via cd .. brings the shell into scope for multi-directory file operations. The move utility (mv) securely ports the target file system link from the tools/ path straight into the target Backup/ storage stack while executing a systematic filename overhaul concurrently. This is validated completely via ls Backup.



□ Step 4: Directory Cleanup Operations

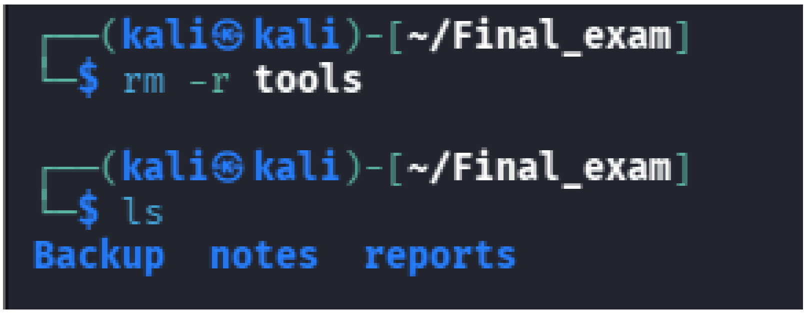
Requirement: Delete the tools directory from the workspace.

Executed Terminal Commands:

$ rm -r tools

$ ls

Technical Explanation: Because the target tools folder contains persistent objects (systeminfo.sh), the recursive configuration flag -r is passed to the remove binary command rm. This enforces a deep cleaning sequence that fully wipes the development layout directory. Running the subsequent list structure check ls explicitly verifies that only Backup, notes, and reports remain active.



□ Step 5: User Provisioning & Privilege Escalation

Requirement: Create a new user named john, set a password for the user, and add the user to the sudo group.

Executed Terminal Commands:

$ sudo adduser john

$ sudo usermod -aG sudo john

$ groups john

Technical Explanation: The high-level interactive deployment command sudo adduser john handles the complete user lifecycle including secure home directory generation and initialization vectors. The administrative modifications syntax usermod -aG sudo appends user profiles smoothly into the elevated security system group without breaking standard permissions. Validation is completely confirmed by the terminal mapping array in the final groups john execution output.

```
┌──(kali㉿kali)-[~/Final_exam]
└─$ sudo adduser john
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for john
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y

┌──(kali㉿kali)-[~/Final_exam]
└─$ sudo usermod -aG sudo john

┌──(kali㉿kali)-[~/Final_exam]
└─$ groups john
john : john sudo users
```

Scenario No: 2 | Service Enumeration & Vulnerability Identification

Task Objective: Perform systematic reconnaissance and active infrastructure service enumeration against the target host machine to uncover exposed communication vectors, determine precise banner/software versions, map out the directory structure of the running web server application, and pinpoint potential points of entry.

□ Step 1: Initial Host Connectivity Validation (Pre-Reconnaissance)

Requirement: Confirm direct network layer reachability and baseline path response to the target system before deploying advanced scanning suites.

Executed Terminal Commands:

$ ping -c 4 10.49.173.86

Technical Explanation: The ICMP echo request tool ping was initiated with a total count parameter of -c 4 targeting the explicit IP address 10.49.173.86. The output displays a 0% packet loss return vector across all 4 segments with an average Round-Trip Time (RTT) metric of roughly 1.082 ms. This establishes a highly stable and operational routing path directly to the laboratory machine.

```
root@ip-10-49-131-161:~# ping -c 4 10.49.173.86
PING 10.49.173.86 (10.49.173.86) 56(84) bytes of data.
64 bytes from 10.49.173.86: icmp_seq=1 ttl=64 time=2.32 ms
64 bytes from 10.49.173.86: icmp_seq=2 ttl=64 time=0.693 ms
64 bytes from 10.49.173.86: icmp_seq=3 ttl=64 time=0.991 ms
64 bytes from 10.49.173.86: icmp_seq=4 ttl=64 time=0.330 ms

--- 10.49.173.86 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3024ms
rtt min/avg/max/mdev = 0.330/1.082/2.317/0.750 ms
```

◻ Step 2: Target Port Validation Scanning via Single-Port TCP Probe

Requirement: Verify high-level ingress filter behaviors on typical communication protocols using raw socket connection parameters.

Executed Terminal Commands:

$ sudo hping3 -S -p 80 -c 4 10.49.173.86

Technical Explanation: The advanced network packet construction utility hping3 was run with administrative rights using the TCP SYN signal flag (-S) targeting endpoint Port 80 (-p 80) across a sequence loop of 4 total transmissions (-c 4). The incoming sequence returned explicitly marked flags acknowledging an RA (Reset-Acknowledgment) response structure across a 0% packet loss layer. This structural state safely reports an active endpoint machine tracking incoming framework parameters.

```
root@ip-10-49-131-161:~# sudo hping3 -S -p 80 -c 4 10.49.173.86
sudo: unable to resolve host ip-10-49-131-161: Name or service not known
HPING 10.49.173.86 (ens5 10.49.173.86): S set, 40 headers + 0 data bytes
len=40 ip=10.49.173.86 ttl=64 DF id=0 sport=80 flags=RA seq=0 win=0 rtt=7.6 ms
len=40 ip=10.49.173.86 ttl=64 DF id=0 sport=80 flags=RA seq=1 win=0 rtt=3.5 ms
len=40 ip=10.49.173.86 ttl=64 DF id=0 sport=80 flags=RA seq=2 win=0 rtt=3.4 ms
len=40 ip=10.49.173.86 ttl=64 DF id=0 sport=80 flags=RA seq=3 win=0 rtt=7.4 ms

--- 10.49.173.86 hping statistic ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 3.4/5.5/7.6 ms
```

◻ Step 3: Complete Port Boundary Active Identification Mapping

Requirement: Perform a full port discovery process covering all potential address flags across the network stack framework.

Executed Terminal Commands:

$ sudo nmap -p- --min-rate 5000 -T4 10.49.173.86

Technical Explanation: An aggressive global scan parameter nmap -p- was called to run over the complete logical array boundary (spanning all 65,535 possible communication endpoints). To scale scanning operational throughput without dropping network precision, the execution properties enforced a processing baseline threshold set at --min-rate 5000 under an accelerated execution timing vector (-T4). The output successfully isolated exactly 6 open listening sockets across the host interface boundary.

```
root@ip-10-49-131-161:~# sudo nmap -p- --min-rate 5000 -T4 10.49.173.86
sudo: unable to resolve host ip-10-49-131-161: Name or service not known
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-17 14:11 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.49.173.86
Host is up (0.0045s latency).
Not shown: 65529 closed ports
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
3128/tcp open  squid-http
3333/tcp open  dec-notes
MAC Address: 0A:82:82:10:67:53 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 4.39 seconds
```

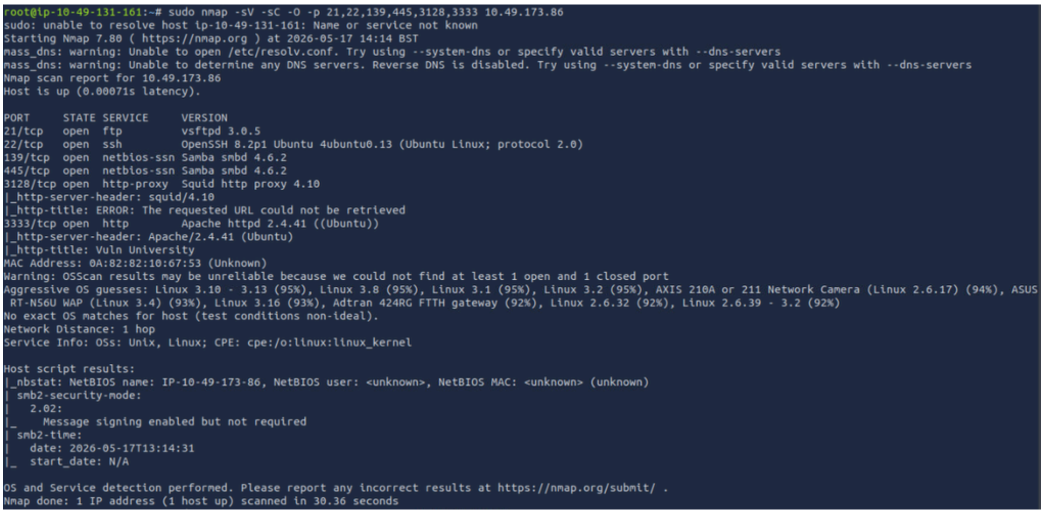
◻ Step 4: Full Service Version Fingerprinting & Core OS Analysis

Requirement: Inspect the identified open ports to determine exact application versions, service identities, and underlying system configurations.

Executed Terminal Commands:

$ sudo nmap -sV -sC -O -p 21,22,139,445,3128,3333 10.49.173.86

Technical Explanation: A comprehensive version enumeration profile scan was executed by targeting only the validated listening ports discovered in the previous phase. The configuration combined service version discovery (-sV), execution of standard Nmap Scripting Engine engine queries (-sC), and active operating system identification probes (-O).



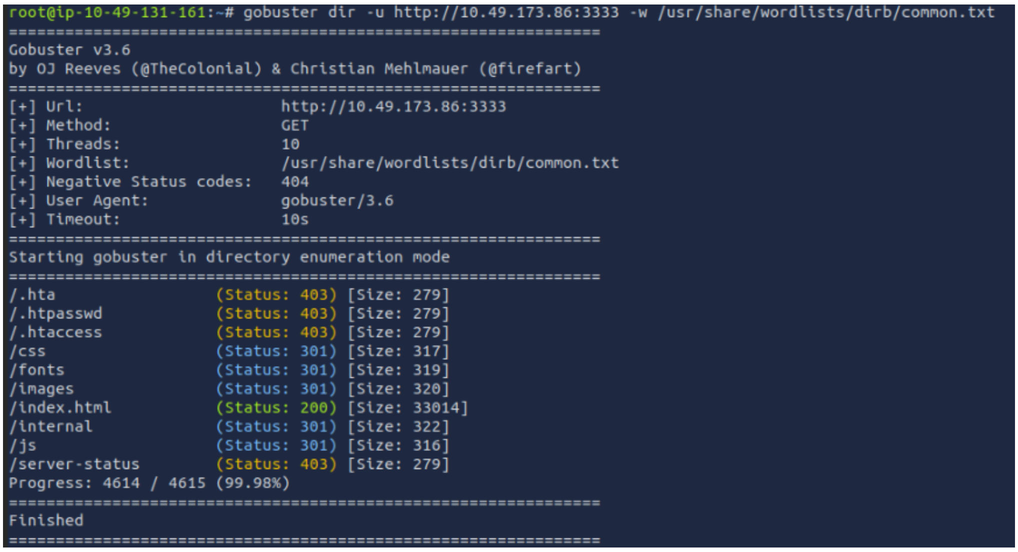
```
root@ip-10-49-131-161:~# sudo nmap -sV -sC -O -p 21,22,139,445,3128,3333 10.49.173.86
sudo: unable to resolve host ip-10-49-131-161: Name or service not known
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-17 14:14 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.49.173.86
Host is up (0.00071s latency).

PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 3.0.5
22/tcp   open  ssh         OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
139/tcp  open  netbios-ssn Samba smbd 4.6.2
445/tcp  open  netbios-ssn Samba smbd 4.6.2
3128/tcp open  http-proxy  Squid http proxy 4.10
|_http-server-header: squid/4.10
|_http-title: ERROR: The requested URL could not be retrieved
3333/tcp open  http        Apache httpd 2.4.41 ((Ubuntu))
|_http-server-header: Apache/2.4.41 (Ubuntu)
|_http-title: Vuln University
MAC Address: 0A:82:82:10:67:53 (Unknown)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 3.10 - 3.13 (95%), Linux 3.8 (95%), Linux 3.1 (95%), Linux 3.2 (95%), AXIS 210A or 211 Network Camera (Linux 2.6.17) (94%), ASUS
 RT-N56U WAP (Linux 3.4) (93%), Linux 3.16 (93%), Adtran 424RG FTTH gateway (92%), Linux 2.6.32 (92%), Linux 2.6.39 - 3.2 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_nbstat: NetBIOS name: IP-10-49-173-86, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb2-security-mode:
|   2.02:
|_    Message signing enabled but not required
| smb2-time:
|   date: 2026-05-17T13:14:31
|_  start_date: N/A

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 30.36 seconds
```

◻ Step 5: Web Application Hidden Directory Enumeration

Requirement: Enumerate internal pathways and hidden application indexes within the active Apache application instance running on Port 3333.

Executed Terminal Commands:

$ gobuster dir -u http://10.49.173.86:3333 -w /usr/share/wordlists/dirb/common.txt



□ Step 6: Target Entry Vector Structural Inspection via Web Browser

Requirement: Access and analyze the front-end layout structure of the identified hidden /internal web route to locate operational flaws.

Target Path URL Location: http://10.49.173.86:3333/internal/